Financial Time Series
46-929
Homework #5

1. GARCH-in-Mean

The text by Hamilton (p667) states "Finance theory suggests that an asset with a higher perceived risk would pay a higher return on average. For exampe, let Y_t denote the ex-post rate of return on some asset minus the return on a safe alternative asset. Suppose that Y_t is decomposed into a component anticipated by investors at time $t-1$ (denoted by m_t) and a component that was unanticipated (denoted by a_t):

$$Y_t = m_t + a_t.$$

Then the theory suggests that the mean return (m_t) would be related to the variance of the return σ_t^2. In general, the ARCH-M (GARCH-M) called ARCH-in-mean (GARCH-in-mean) is given by:

$$Y_t = \beta_0 + \beta_1 X_t + \delta\sigma_t^2 + a_t,$$

$$a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = 1 + 0.4a_{t-1}^2.$$

Here the $\{\epsilon_t$ are i.i.d. N(0,1) random variables." The ARCH/GARCH-in-mean model has the additional feature that the conditional volatility appears in the mean model. Assume that $\beta_0 = .06$, $\beta_1 = 0.30$, and $\delta = 0.20$, and the process $\{X_t\}$ is the inflation rate during this time period.

a) What is $E(Y_t|X_t = 0.1, a_{t-1} = 0.6)$?

b) What is $Var(Y_t|X_t = 0.1, a_{t-1} = 0.6)$?

c) Is the conditional distribution of Y_t given X_t and a_{t-1} normal? Why or why not?

d) Is the marginal distribution of Y_t normal? Why or why not?

Solution

a)

$$\begin{aligned}
E(Y_t|X_t, a_{t-1}) &= E(\beta_0 + \beta_1 X_t + \delta\sigma_t^2 + a_t|X_t, a_{t-1}) \\
&= \beta_0 + \beta_1 X_t + \delta(1 + .4a_{t-1}^2) + E(a_t|X_t, a_{t-1}) \\
&= \beta_0 + \beta_1 X_t + \delta(1 + .4a_{t-1}^2) + E(\epsilon_t\sqrt{1 + .4a_{t-1}^2}|X_t, a_{t-1}) \\
&= \beta_0 + \beta_1 X_t + \delta(1 + .4a_{t-1}^2) + E(\epsilon_t)E(\sqrt{1 + .4a_{t-1}^2}|X_t, a_{t-1})) \\
&= \beta_0 + \beta_1 X_t + \delta(1 + .4a_{t-1}^2) + 0 \\
&= 0.06 + 0.30(0.1) + 0.20(1 + .4 * 0.6^2) \\
&= .3188
\end{aligned}$$

b)

$$\begin{aligned}
Var(Y_t|aX_t, a_{t-1}) &= Var(\beta_0 + \beta_1 X_t + \delta\sigma_t^2 + a_t|X_t, a_{t-1}) \\
&= Var(\delta\sigma_t + a_t|X_t, a_{t-1}) \\
&= Var(a_t|a_{t-1})
\end{aligned}$$

$$
\begin{aligned}
&= Var(\epsilon_t \sigma_t | a_{t-1}) - 0 \\
&= E(\epsilon_t^2(\omega + .4a_{t-1}^2)|a_{t-1}) \\
&= 1 * (\omega + .4a_{t-1}^2) \\
&= 1 + .4 * .6 * .6 \\
&= 1.144
\end{aligned}
$$

c) Note that conditional on X_t and a_{t-1}, $\beta_0 + \beta_1 X_t + \delta\sigma_t$ is constant, since $\sigma_t = \sqrt{\omega + .4a_{t-1}^2}$. Thus the conditional distribution of Y_t is a constant + $\sqrt{\omega + .4a_{t-1}^2}\epsilon_t$ which has a normal distribution because conditionally on a_{t-1}, σ_t is a constant.

d) First, the probability distribution of the inflation rate, X_t, is not given. Consequently, one cannot conclude that the distribution of Y_t will be normal. But going further, recall that the kurtosis of a_t is larger than 3, so the noise term in the definition of the Y_t model is not normal, and so unconditionally Y_t will not be normal.

2. Download stock data Merck (ticker = MRK) from 1/1/2011 to 4/23/2021. Run through a complete analysis of time series for the log-returns of this data series.

   (a) Plot the log-returns and analyze the mean model. Check for stationarity using the adjusted Dickey Fuller test. Make any necessary transformations to make the series stationary.

   (b) Look at the ACF and PACF plots and test for any significant autocorrelation using the Ljung-Box test. Report your results.

   (c) If there are any significant autocorrelations, find a good ARMA model for these data.

   (d) Test for any ARCH effects using the Ljung-Box test and the Engle test.

   (e) If there are any significant ARCH effects, fit a GARCH model to the data.

   (f) Using the data up to 4/1/21, forecast the volatility going forward to 4/23/2021.

   (g) Suppose you own 100 shares of MRK on 4/1/2021. What is a 99% confidence interval for the price of MRK on 4/23/2020 based on the models you have developed?

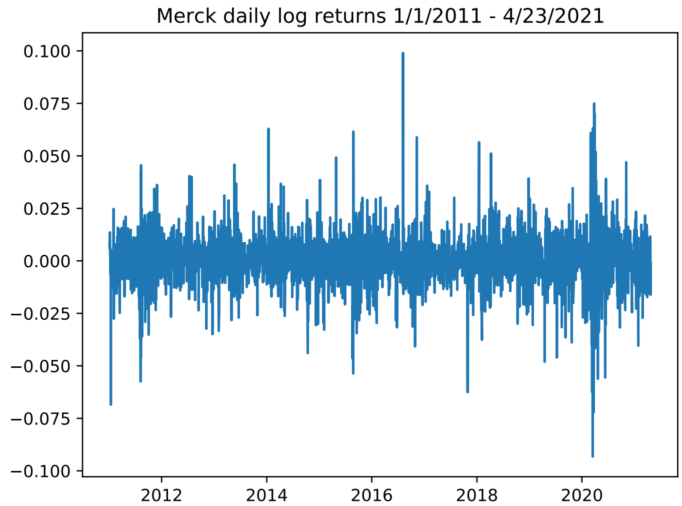
Solution

a,b)
```
import pandas as pd
import pandas_datareader as pdr
import numpy as np
import statsmodels.api as smm
import statsmodels.formula.api as smf
import statsmodels.tsa.stattools as smt
import scipy.stats as scs
import matplotlib.pyplot as plt
from statsmodels.stats.diagnostic import het_arch
from arch import arch_model
from scipy.stats import norm, t

start = '2011-01-01'
end = '2021-4-23'

price_data = pdr.get_data_yahoo('mrk',start,end)
price_data_adjclose = price_data['Adj Close']
logret= np.log(price_data_adjclose/price_data_adjclose.shift(1)).dropna()
```

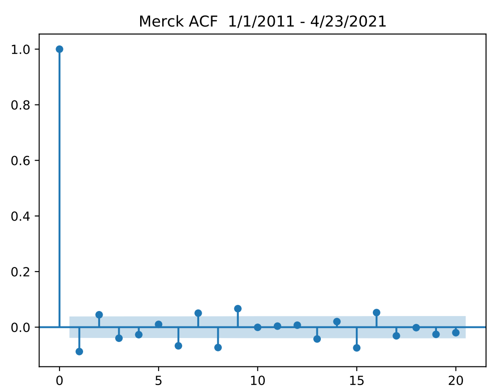
```
smt.adfuller(logret)
```

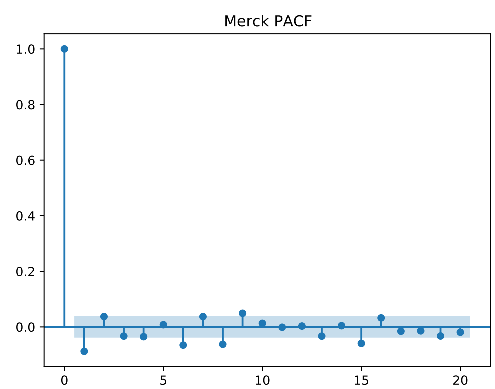



```
smt.adfuller(logret)
(-13.533680410315649, 2.5922903849253684e-25, 15, 2577, {'1%': -3.4328900953577164, '5
```

The Dickey Fuller test clearly rejects the null hypothesis (unit root) statistic of -13.53 gives p-value of 0. - thus the series is stationary and no differencing is required.

```
smm.graphics.tsa.plot_acf(logret,lags = 20,alpha = .05)
smm.graphics.tsa.plot_pacf(logret,lags = 20,alpha = .05)
```






```
smm.stats.diagnostic.acorr_ljungbox(logret,10)
```

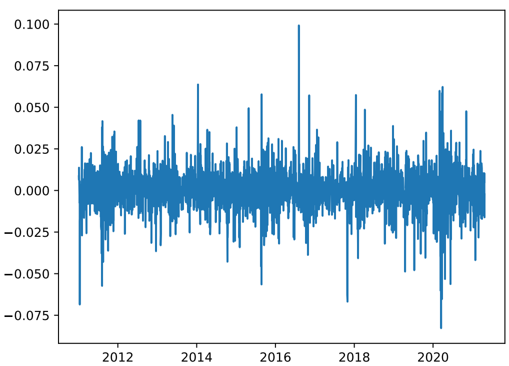
(array([19.96204106, 25.15249829, 29.24546171, 31.1295321 , 31.39742385, 43.1001994 , 49.743647 , 63.67265246, 75.24745203, 75.24832642]), array([7.89949054e-06, 3.45306213e-06, 1.98862024e-06, 2.88066860e-06, 7.81803020e-06, 1.11437349e-07, 1.62190664e-08, 8.82678962e-11, 1.41212742e-12, 4.25660761e-12]))
Clearly there are significant autocorrelations with very tiny p-values.

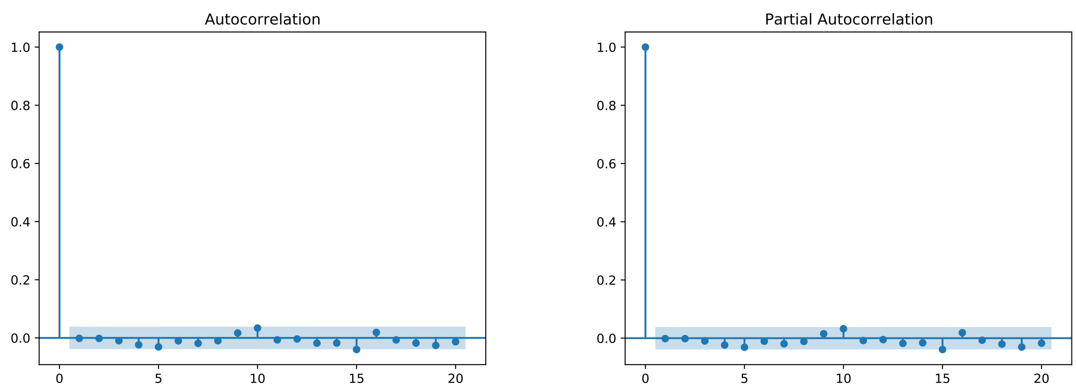
c) Using the ARMA order select command:

```
res_aic = smm.tsa.arma_order_select_ic(logret,ic='aic', trend ='nc')
```

```
print(res_aic.aic_min_order)
res_bic = smm.tsa.arma_order_select_ic(logret,ic='bic', trend ='nc')
print(res_bic.bic_min_order)
```

Both the AIC and the BIC as select an ARMA(4,2) model.





This shows that the temporal effects have been modeled. However, we need to check for ARCH effects.

d) Both the Ljung-Box test and the Engle test confirm the existence of ARCH effect. First, the Ljung-Box test:

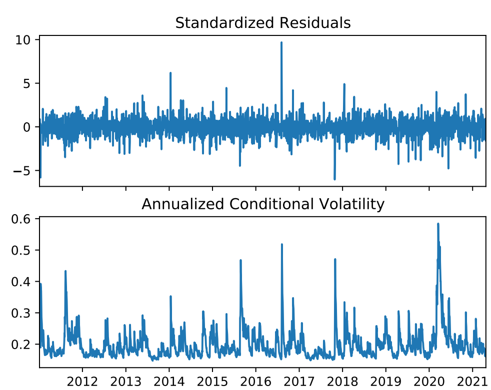
```
smm.stats.diagnostic.acorr_ljungbox(model.resid**2,lags = 10)

(array([ 68.06280581, 138.85000818, 180.98320213, 195.94145383,
       269.25361566, 276.10938136, 300.03560939, 359.01534258,
       366.77811932, 433.52745577]), array([1.58369963e-16, 7.06486418e-31, 5.40924005
       4.04772673e-56, 1.06898937e-56, 5.94463492e-61, 1.07692330e-72,
       1.65816714e-73, 6.79978709e-87]))
```

Next the Engle test for ARCH Effects:
print(het_arch(model_aic.resid**2, 10))
(62.38167053966375, 1.2803379499676652e-09, 6.365329282611634, 9.86718913699239e-10)
Both given very strong evidence for the presence of ARCH effects

e) am = arch_model(model.resid)
res.summary()

```
                     Constant Mean - GARCH Model Results
==============================================================================
Dep. Variable:                      None   R-squared:                       0.000
Mean Model:                Constant Mean   Adj. R-squared:                  0.000
Vol Model:                         GARCH   Log-Likelihood:                7758.60
Distribution:                     Normal   AIC:                          -15509.2
Method:               Maximum Likelihood   BIC:                          -15485.8
                                           No. Observations:                 2593
Date:                   Sat, May 01 2021   Df Residuals:                     2592
Time:                           23:28:14   Df Model:                            1
                                 Mean Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
mu          7.9712e-04  2.447e-04      3.258  1.122e-03 [3.176e-04,1.277e-03]
                              Volatility Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
omega       1.6482e-05  2.853e-12  5.777e+06      0.000 [1.648e-05,1.648e-05]
alpha[1]        0.1000  2.787e-02      3.589  3.326e-04   [4.538e-02,  0.155]
beta[1]         0.8000  2.760e-02     28.981 1.155e-184     [  0.746,  0.854]
============================================================================
```



To check if the ARCH effects have been accounted for, we run the Engle test
statsmodels.stats.diagnostic.het_arch(std_resid)

(7.459031674551876, 0.9999633193158198, 0.2641502767269545, 0.9999651407360753)

The p-values for the Engle test and the F test are essentially 1.0 giving strong assurance that the ARCH effects have been addressed by the GARCH(1,1) model with ω nearly 0, $\alpha = 0.1$ and $\beta = 0.8$. The persistence parameter $\lambda = \alpha + \beta = .9$

One can also run the Ljung-Box test on the squared standardized residuals. The results also give very strong evidence that the standardized squared residuals are white noise indicating that the ARCH effects have been fully modeled.

statsmodels.stats.diagnostic.acorr_ljungbox(std_resid**2, lags = 15)

```
(array([ 0.40994019,  0.46671663,  0.54511048,  1.99915097,  2.12803247,
        2.69836204,  2.9423119 ,  3.03226827,  3.75596718,  3.79287652,
        3.92384888,  3.94044916,  3.95461641,  4.18713149,  4.22266757,
        4.47974413,  4.53199072,  5.10715822,  5.21170996,  5.29994968,
        5.36275621,  5.40454748,  6.73665439,  6.84335645,  6.94854879,
        7.83200324,  8.17951689,  8.1935596 , 10.12928198, 10.12929825]), array([0.52
       0.84564073, 0.89028705, 0.93231623, 0.92672764, 0.9562073 ,
       0.97209185, 0.98448713, 0.99166462, 0.99423118, 0.99691202,
       0.99779409, 0.99881503, 0.99868321, 0.99923142, 0.99956528,
       0.99976552, 0.99987908, 0.99960314, 0.99976362, 0.99986044,
       0.99977584, 0.99981874, 0.99990236, 0.99954509, 0.99974142]))
```

f) This part of the problem involves using data up to 4/1/21 to forecast volatility from 4/2/21 to 4/23/21. We need to be careful to use only the data up to 4/1/21 and refit the models using the slightly smaller dataset to avoid "look ahead bias."

We execute the same commands as before on the smaller data set (not shown here): (getting variables logret1, model1, am1 and res1). The mean model is still best fit by an ARMA(4,2) model (AR IMA(4,0,2)). The GARCH volatility is also fit by a GARCH(1,1) model with nearly identical parameter values.

```
            Volatility
Date
2021-04-02    0.011532
2021-04-05    0.011680
2021-04-06    0.011812
2021-04-07    0.011929
2021-04-08    0.012034
2021-04-09    0.012127
2021-04-12    0.012211
2021-04-13    0.012285
2021-04-14    0.012352
2021-04-15    0.012412
2021-04-16    0.012466
2021-04-19    0.012514
2021-04-20    0.012557
2021-43-21    0.012595
2021-04-22    0.012630
```

g) Recall that problem 2g was cancelled.